

$$\textcircled{9} \lim_{x \rightarrow 0} \frac{\sin 2x}{e^{2x} - e^{-2x}} \leftarrow \frac{0}{0}$$

$$\underline{\underline{L'Hosp}} \quad \lim_{x \rightarrow 0} \frac{2\cos 2x}{2e^{2x} + 2e^{-2x}} = \frac{1}{2}$$

$$\textcircled{10} \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{1-\cos x}} \quad \begin{matrix} \infty \\ 1^\infty \end{matrix}$$

$$\lim_{x \rightarrow 0} \frac{1}{1-\cos x} \cdot \ln \frac{\sin x}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\ln \sin x - \ln x}{1 - \cos x}$$

$$\underline{\underline{L'Hosp}} \quad \lim_{x \rightarrow 0} \frac{\frac{1}{\sin x} \cdot \cos x - \frac{1}{x}}{\sin x}$$

$$= \lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{x \sin^2 x}$$

$$= \lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{x^3}$$

$$\underline{\underline{L'Hosp}} \quad \lim_{x \rightarrow 0} \frac{\cos x - x \sin x - \cos x}{3x^2}$$

$$= \lim_{x \rightarrow 0} \frac{-x \sin x}{3x^2} = -\frac{1}{3}$$

$$\Rightarrow \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{x}} = e^{-\frac{1}{3}}$$

Way 2 : $\lim_{x \rightarrow 0} \frac{\ln \left(\frac{\sin x}{x} \right) \rightarrow 1}{1 - \cos x}$

$\ln(1+x) \sim x \quad (x \rightarrow 0)$

$\ln x \sim x-1 \quad (x \rightarrow 1)$

$$= \lim_{x \rightarrow 0} \frac{\ln \left(1 + \frac{\sin x}{x} - 1 \right)}{\frac{1}{2} x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{\sin x}{x} - 1}{\frac{1}{2} x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x - x}{\frac{1}{2} x^3}$$

$$= \lim_{x \rightarrow 0} \frac{\cos x - 1}{\frac{3}{2} x^2}$$

$$= \lim_{x \rightarrow 0} \frac{-\frac{1}{2}x^2}{\frac{3}{2}x^2}$$

$$= -\frac{1}{3}$$

$$\begin{matrix} x \rightarrow 0 \\ e^{3x} - 1 \rightarrow 0 \end{matrix}$$

$$(11) \lim_{x \rightarrow 0} \frac{\sin^2 5x \tan^3 2x \arcsin(e^{3x}-1)}{x^2 \ln^3(2x+1) \arctan x}$$

$$= \lim_{x \rightarrow 0} \frac{(5x)^2 (2x)^3 \cdot 3x}{x^2 (2x)^3 \cdot x}$$

$$= 75$$

$$(12) F(x) = \int_{x^2}^4 x f(t) dt$$

$$\text{求 } F'(2).$$

$$\frac{d \left(\int_0^{g(x)} f(t) dt \right)}{dx} = f(g(x)) \cdot g'(x)$$

$$\frac{d \left(\int_0^{x^2} f(t) dt \right)}{dx} = f(x^2) \cdot (x^2)'$$

$$= 2x f(x^2)$$

$$\frac{d \left(\int_{\underline{x^2}}^0 f(t) dt \right)}{dx} =$$

$$= \frac{d \left(- \int_0^{x^2} f(t) dt \right)}{dx} = -2x f(x^2)$$

$$\frac{d \left(\int_{x^2}^{x^3} f(t) dt \right)}{dx} = \frac{d \left(\int_0^{x^3} f(t) dt - \int_0^{x^2} f(t) dt \right)}{dx}$$

$$= 3x^2 f(x^3) - 2x f(x^2).$$

$$\frac{d \left(\int_0^x x f(t) dt \right)}{dx}$$

$$F(x) = \int_0^x 2xt \, dt = 2x \int_0^x t \, dt = x^3$$

$$\text{th } F'(x) = 2x \cdot x = 2x^2 \cdot x.$$

x

$$F(x) = 2x \left[\int_0^x f(t) dt \right]$$

$$= 2 \cdot \int_0^x f(t) dt + 2x f(x) \quad \leftarrow f(x) = x$$

$$= 3x^2$$

$$(2) F(x) = \int_{-1}^{x^2} \sin t \, dt$$

$$F'(x) = \sin x^2 \cdot (x^2)'$$

(3)

$$F(x) = \int_{x^2}^4 x f(t) \, dt$$

$$\text{find } F'(x)$$

$$F(x) = x \left[\int_{x^2}^4 f(t) \, dt \right]$$

$$F'(x) = \int_{x^2}^4 f(t) \, dt + x \cdot [-(x^2)' \cdot f(x^2)]$$

$$= \int_{x^2}^4 f(t) \, dt - 2x^2 f(x^2)$$

$$F'(2) = \underbrace{\int_4^4 f(t) dt}_{\downarrow 0} - 8f(4)$$

$$= -8f(4)$$

$$F'(1) = \int_1^4 f(t) dt - 2f(1)$$

$$F(x) = \underbrace{x}_{g(x)} \cdot \underbrace{\int_0^x f(t) dt}_{h(x)}$$

$$= g'(x) \cdot h(x) + g(x) \cdot h'(x)$$

$$(13) \int x \arctan x \, dx$$

$$= \int x \cdot (\arctan x \, dx) \quad \times$$

$$= \int (\arctan x) \underline{(x dx)}$$

$$= \underline{\frac{1}{2} \int \arctan x \, d x^2}$$

$$= \frac{1}{2} x^2 \arctan x - \frac{1}{2} \int x^2 d \arctan x$$

$$= \frac{1}{2} x^2 \arctan x - \frac{1}{2} \int \frac{x^2}{1+x^2} dx$$

$$= \frac{1}{2} x^2 \arctan x - \frac{1}{2} \int \left(1 - \frac{1}{1+x^2}\right) dx$$

$$= \frac{1}{2} x^2 \arctan x - \frac{1}{2} x + \frac{1}{2} \arctan x$$

$$= F(x)$$

$$\int_0^1 x \arctan x \, dx = F(1) - F(0).$$

$$= \frac{1}{2} \cdot 1 \cdot \arctan 1$$

$$- \frac{1}{2} + \frac{1}{2} \arctan 1$$

$$- (0 - 0 + 0).$$

$$= \arctan 1 - \frac{1}{2}$$

$$= \frac{\pi}{4} - \frac{1}{2}$$

(14) . (15) 下次

$$(16) f(x) = x^x \quad (x > 0)$$

解方程: $f(x) = f'(x)$.

$$x^x = x^x + x^x \ln x$$

$$\Rightarrow \underline{x^x \ln x} = 0$$

$$\Rightarrow x = 1$$

⑪ $f(x) = (x^2 - 1)^2 - \frac{1}{2} \sqrt{1 - x^2}$ 的 ~~最~~ 值

求

定义域: $[-1, 1]$

maximum
and minimum

Way 1: $f'(x) = 2(x^2 - 1) \cdot \underline{(x^2 - 1)'} - \frac{1}{2} \cdot \frac{1}{2}(1 - x^2)^{-\frac{1}{2}} \cdot \underline{(1 - x^2)'}^{-2x}$

$$= 4x(x^2 - 1) + \frac{1}{2} \frac{x}{\sqrt{1 - x^2}}$$

$$= \frac{4x \cdot (x^2-1) \cdot 2\sqrt{1-x^2} + x}{2\sqrt{1-x^2}} \geq 0$$

$$= \frac{8x(1-x^2)^{\frac{3}{2}} \cdot (-1) + x}{2\sqrt{1-x^2}}$$

$$= \frac{\boxed{x} \left(-8(1-x^2)^{\frac{3}{2}} + 1 \right)}{2\sqrt{1-x^2}} \geq 0$$

$$1 - 8(1-x^2)^{\frac{3}{2}} \geq 0 \quad ?$$

$$\Leftrightarrow 1 \geq 8(1-x^2)^{\frac{3}{2}}$$

$$\Leftrightarrow \frac{1}{8} \geq (1-x^2)^{\frac{3}{2}} = (\sqrt{1-x^2})^3$$

$$\Leftrightarrow \left(\frac{1}{8}\right)^{\frac{1}{3}} \geq \sqrt{1-x^2}$$

$$\Leftrightarrow \frac{1}{2} \geq \sqrt{1-x^2}$$

$$\Leftrightarrow \frac{1}{4} \geq 1-x^2$$

$$\Leftrightarrow \frac{x^2 \geq \frac{3}{4}}{\quad}$$

$$x \geq \frac{\sqrt{3}}{2} \text{ or } x \leq -\frac{\sqrt{3}}{2}$$

$$-1 \quad -\frac{\sqrt{3}}{2} \quad 0 \quad \frac{\sqrt{3}}{2} \quad 1$$

$$\underline{x \in (-1, -\frac{\sqrt{3}}{2})}$$

$$x^2 \geq \frac{3}{4}, \quad x \leq 0$$

$$\underline{(-1, -\frac{\sqrt{3}}{2})}$$

$$\underline{(-\frac{\sqrt{3}}{2}, 0)}$$

$$\underline{(0, \frac{\sqrt{3}}{2})} \quad \underline{(\frac{\sqrt{3}}{2}, 1)}$$

$f'(x)$

-

+

-

+

$f(x)$

↓

↑

↓

↗

$$f(x) \text{ 最大? } f(-\frac{\sqrt{3}}{2}), \quad f(\frac{\sqrt{3}}{2})$$

$$\text{大? } f(0), f(1), f(-1)$$

$$f\left(\frac{\sqrt{3}}{2}\right) = -\frac{3}{16} \leftarrow \frac{10}{16} \text{ 又小}$$

$$f(0) = \frac{1}{2} \leftarrow \frac{10}{16} \text{ 又大}$$

$$f(1) = 0$$

$$(x^2-1)^2 - \frac{1}{2} \sqrt{1-x^2}$$

$$\text{令 } t = 1 - x^2 \in [0, 1]$$

$$u = \sqrt{1-x^2} \in [0, 1]$$

$$g(t) = t^2 - \frac{1}{2} \sqrt{t}$$

$$h(u) = u^4 - \frac{1}{2} u$$

$$g'(t) = 2t - \frac{1}{4} \frac{1}{\sqrt{t}}$$

$$h'(u) = 4u^3 - \frac{1}{2}$$

$$\Rightarrow u \geq \frac{1}{2}, h'(u) \geq 0$$

$$u \leq \frac{1}{2}, h'(u) \leq 0$$

$$g'(t) \geq 0?$$

$$f: h(\frac{1}{2})$$

$$\Leftrightarrow 2t \geq \frac{1}{4\sqrt{t}} \quad t, h(0), h(1)$$

$$\Leftrightarrow 8t^{\frac{3}{2}} \geq 1$$

$$\Leftrightarrow t \geq \frac{1}{4}$$

$$g(t) \quad (0, \frac{1}{4}) \downarrow$$

$$(\frac{1}{4}, 1) \uparrow$$

$$f: g(\frac{1}{4})$$

$$t \quad g(0), g(1)$$